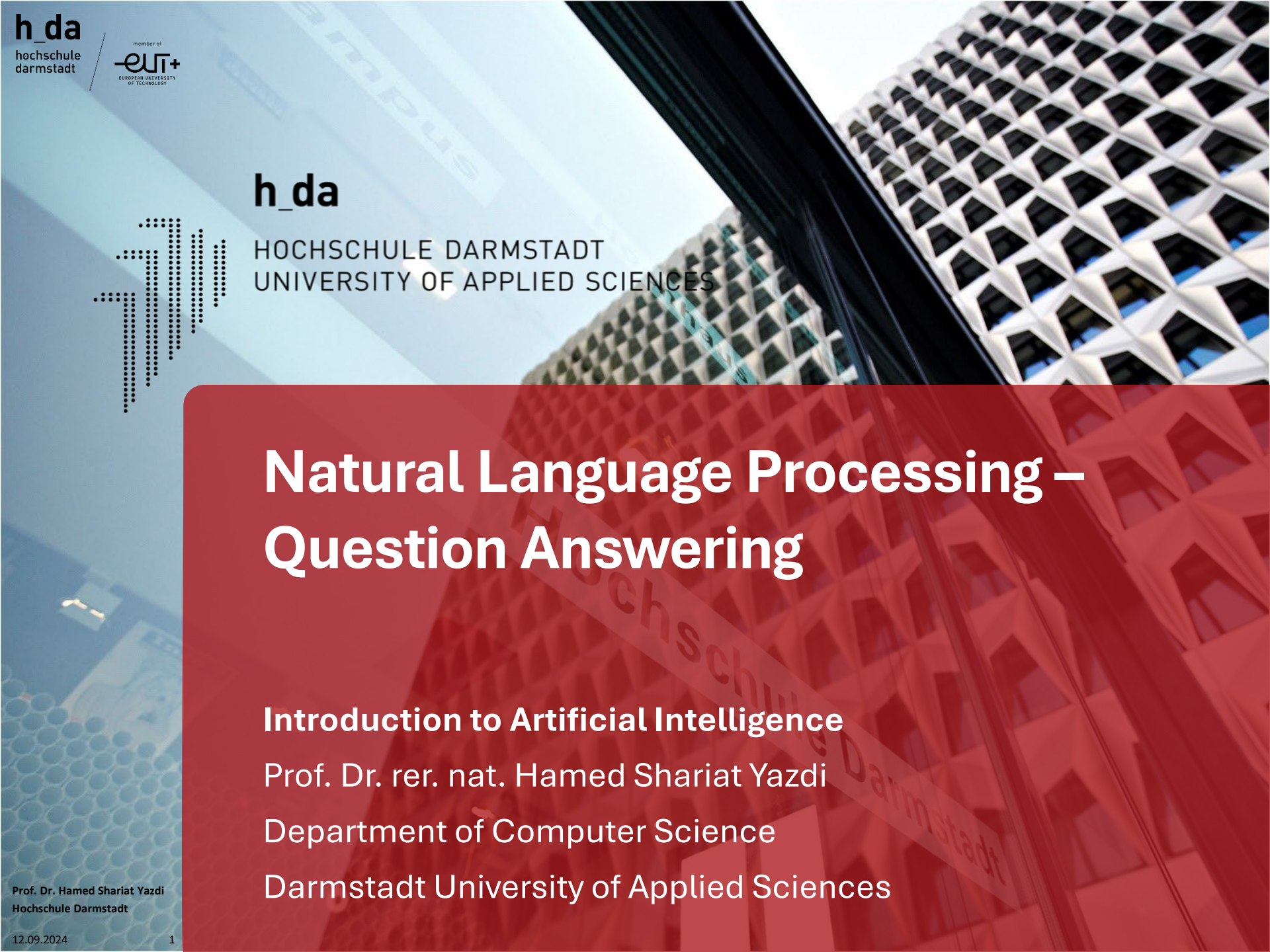


Natural Language Processing – Question Answering

Introduction to Artificial Intelligence

Prof. Dr. rer. nat. Hamed Shariat Yazdi

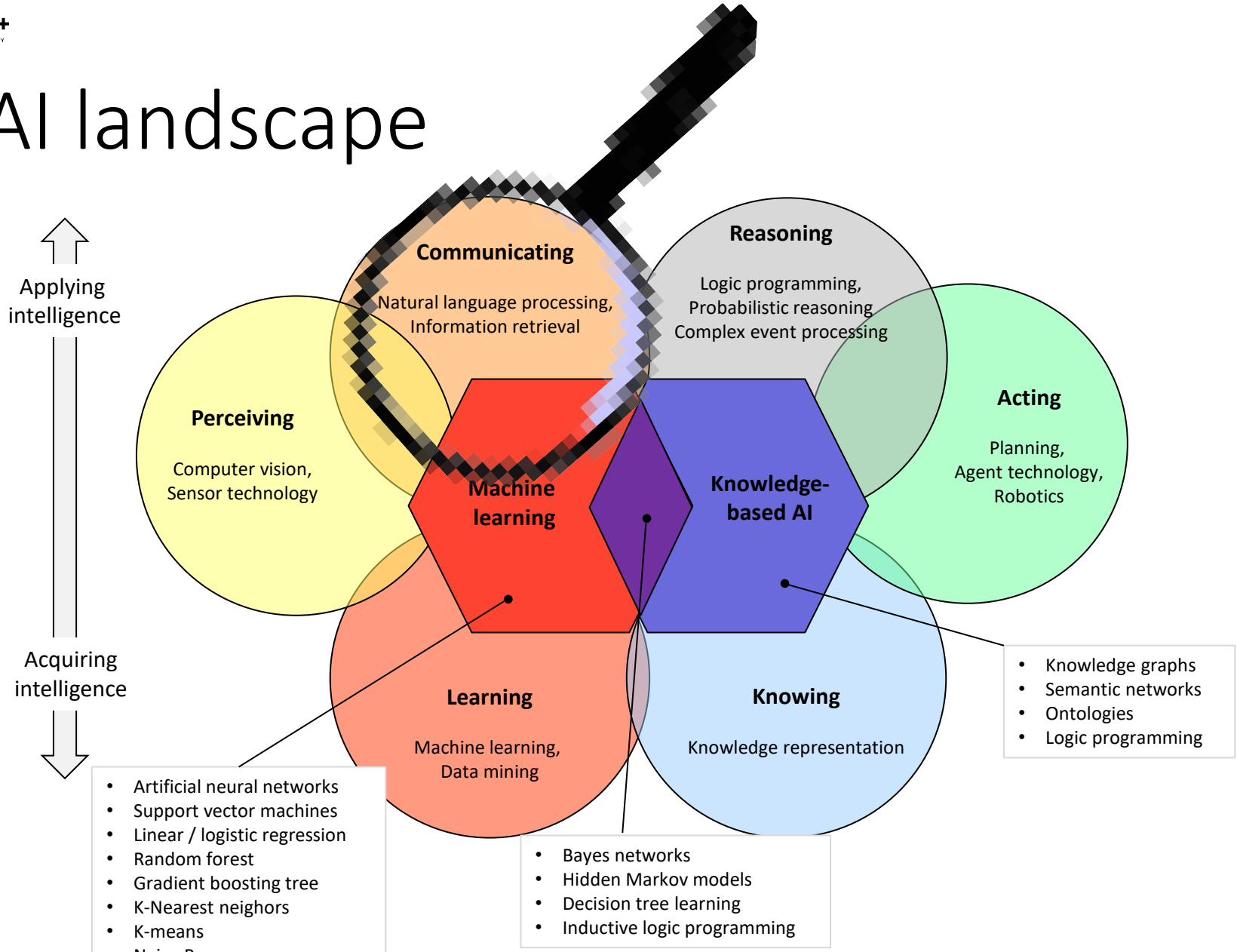
Department of Computer Science

Darmstadt University of Applied Sciences

Agenda

- Overview
- spaCy
- Simple question answering
- Quick check

AI landscape



Inspired by https://www.sigs-datacom.de/order/poster/Landkarte-KI-Poster-AI-2018.php

Prof. Dr. Hamed Shariat Yazdi
Department of Computer Science
Darmstadt Univeristy of Applied Sciences

Natural Language Processing (NLP) Areas

Information Retrieval (IR)	• Retrieving relevant documents (“Document retrieval“ would be more suitable a term)
Text Classification	• Classifying natural language texts, e.g. to detect languages, topics, or sentiment
Information Extraction (IE)	• Understanding written / spoken texts • Building knowledge graphs from texts
Question Answering (QA)	• Answering natural language questions
Machine Translation	• Translating texts between natural languages
Text Generation	• Generate written / spoken texts • Summarizing texts

Prof. Dr. Hamed Shariat Yazdi
Department of Computer Science
Darmstadt Univeristy of Applied Sciences

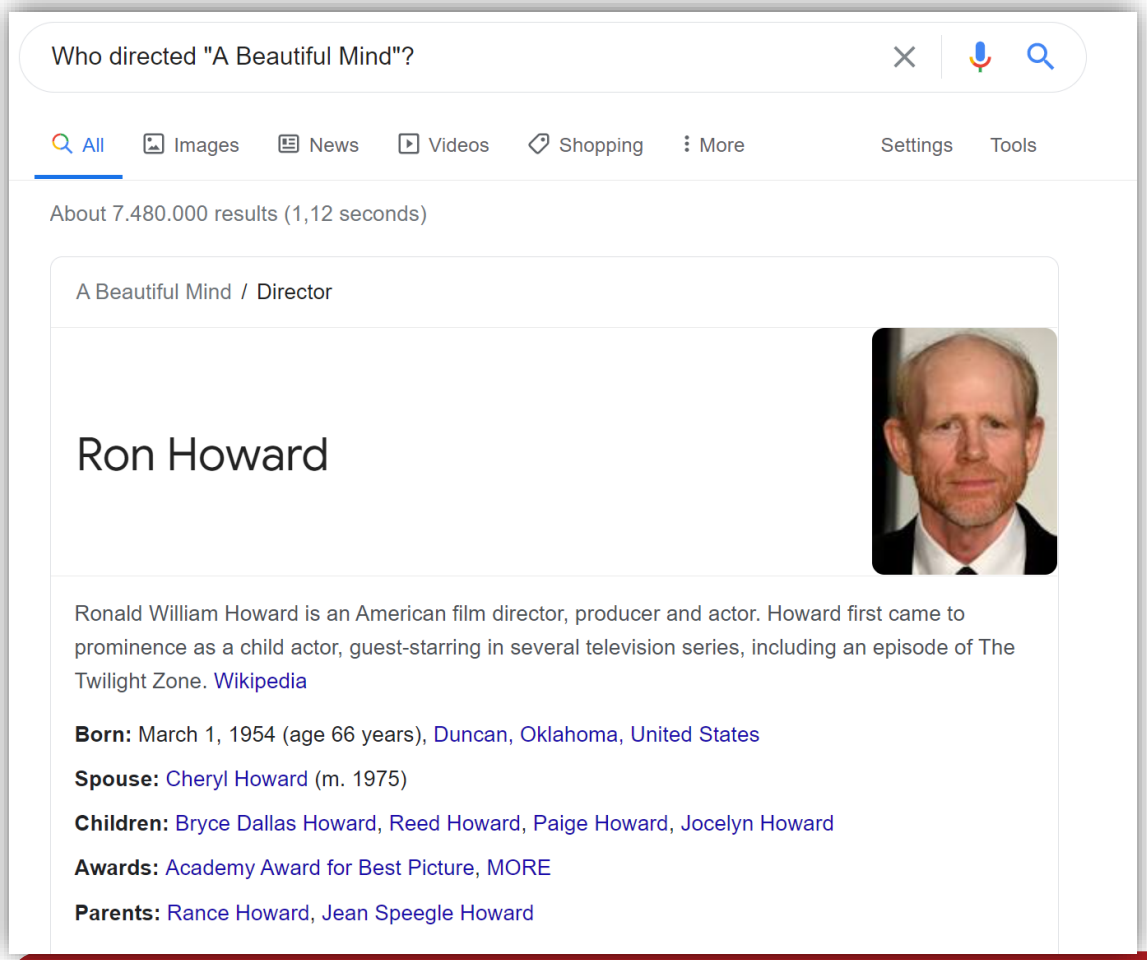
Question answering

- Input: question text → Output: answer text



https://upload.wikimedia.org/wikipedia/commons/5/5c/Amazon_Echo.jpg

Question answering in Google: Who directed "A Beautiful Mind"?



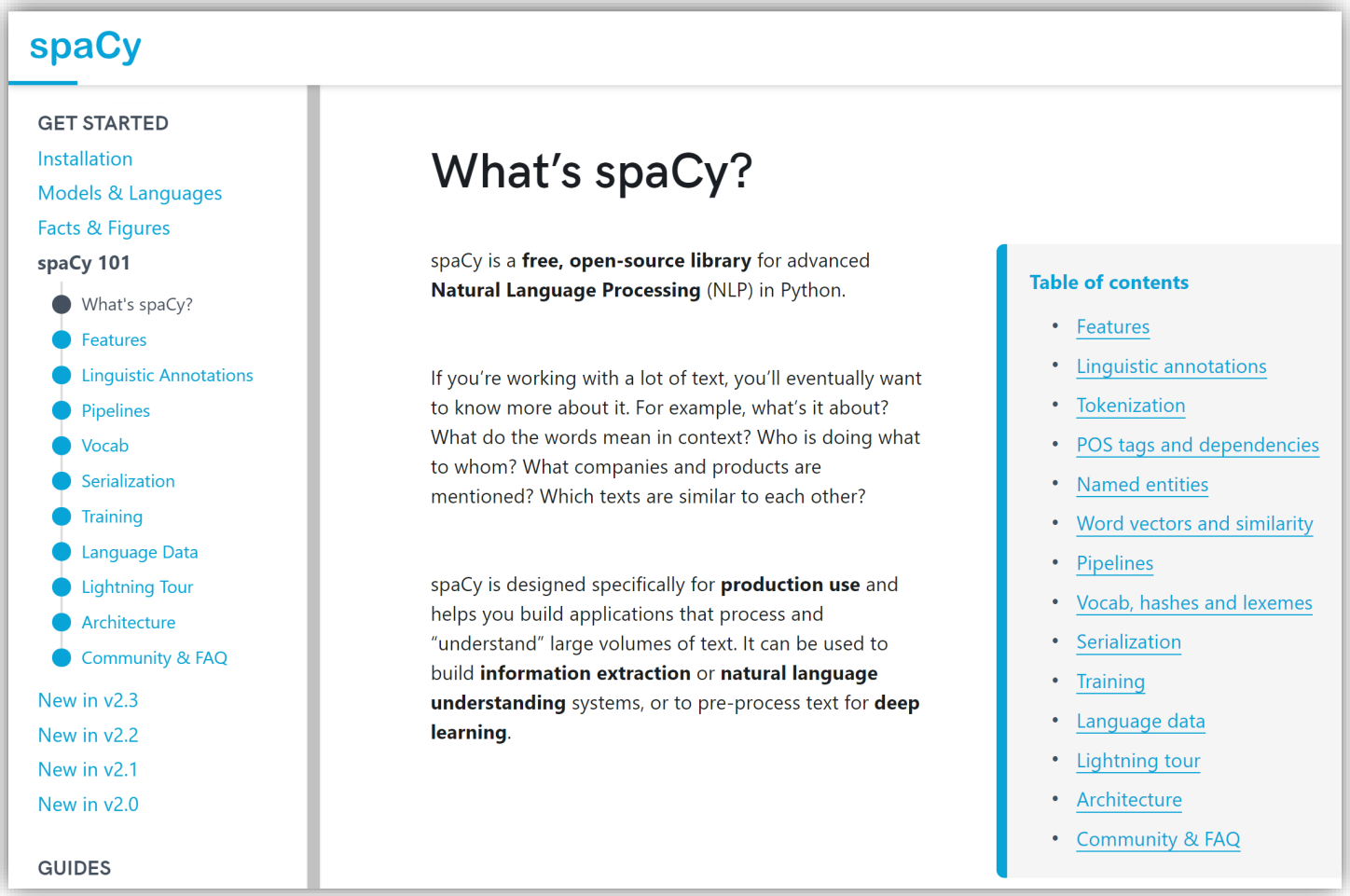
https://www.google.com/search?q=Who+directed+%22A+Beautiful+Mind%22

Prof. Dr. Hamed Shariat Yazdi
Department of Computer Science
Darmstadt Univeristy of Applied Sciences

Agenda

- Overview
- spaCy
- Simple question answering
- Quick check

spaCy: Open source Python library for NLP



https://spacy.io/usage/spacy-101

Prof. Dr. Hamed Shariat Yazdi
Department of Computer Science
Darmstadt Univeristy of Applied Sciences

spaCy functionality

NAME	DESCRIPTION
Tokenization	Segmenting text into words, punctuations marks etc.
Part-of-speech (POS) **Tagging**	Assigning word types to tokens, like verb or noun.
Dependency Parsing	Assigning syntactic dependency labels, describing the relations between individual tokens, like subject or object.
Lemmatization	Assigning the base forms of words. For example, the lemma of "was" is "be", and the lemma of "rats" is "rat".
Sentence Boundary Detection (SBD)	Finding and segmenting individual sentences.
Named Entity Recognition (NER)	Labelling named "real-world" objects, like persons, companies or locations.
Entity Linking (EL)	Disambiguating textual entities to unique identifiers in a Knowledge Base.
Similarity	Comparing words, text spans and documents and how similar they are to each other.
Text Classification	Assigning categories or labels to a whole document, or parts of a document.
Rule-based Matching	Finding sequences of tokens based on their texts and linguistic annotations, similar to regular expressions.
Training	Updating and improving a statistical model's predictions.
Serialization	Saving objects to files or byte strings.

https://spacy.io/usage/spacy-101

Installing models and processing text

Install models and process text

```
python -m spacy download en_core_web_sm
python -m spacy download de_core_news_sm
```

Editable Code spaCy v2.3.0 · Python 3 · via Binder

```
import spacy

nlp = spacy.load("en_core_web_sm")
doc = nlp("Hello, world. Here are two sentences.")
print([t.text for t in doc])

nlp_de = spacy.load("de_core_news_sm")
doc_de = nlp_de("Ich bin ein Berliner.")
print([t.text for t in doc_de])
```

RUN

```
['Hello', ',', 'world', '.', 'Here', 'are', 'two', 'sentences', '.']
['Ich', 'bin', 'ein', 'Berliner', '.']
```

https://spacy.io/usage/spacy-101

Prof. Dr. Hamed Shariat Yazdi
Department of Computer Science
Darmstadt Univeristy of Applied Sciences

Similarity metric

Use at least medium-sized model to get proper simililarity results

```
1  import spacy
2  nlp = spacy.load('en_core_web_md')
3
4  apple = nlp('apple')
5  pear = nlp('pear')
6  bear = nlp('bear')
7
8
9  print('pear <-> apple: ', pear.similarity(apple))
10 print('pear <-> bear: ', pear.similarity(bear))
11
```

```
pear <-> apple:  0.7585767914712307
pear <-> bear:  0.10610448560865217
```

https://spacy.io/usage/spacy-101

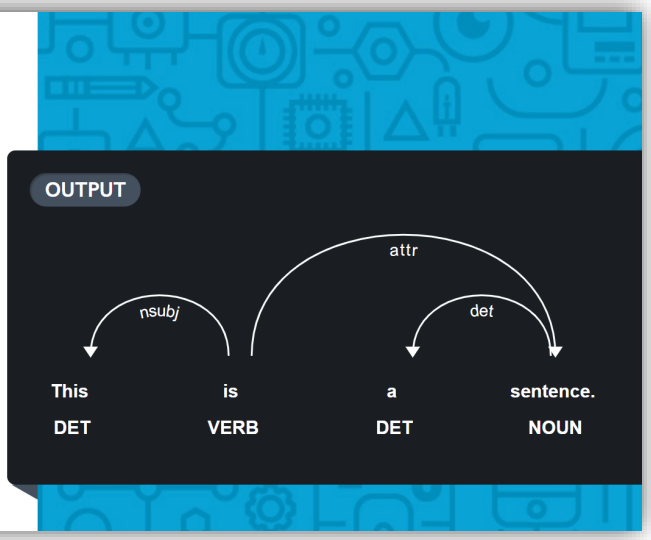
Advanced NLP: Parsing sentences

Visualize a dependency parse and named entities in your browser V2.0 NEEDS MODEL

```
from spacy import displacy

doc_dep = nlp("This is a sentence.")
displacy.serve(doc_dep, style="dep")

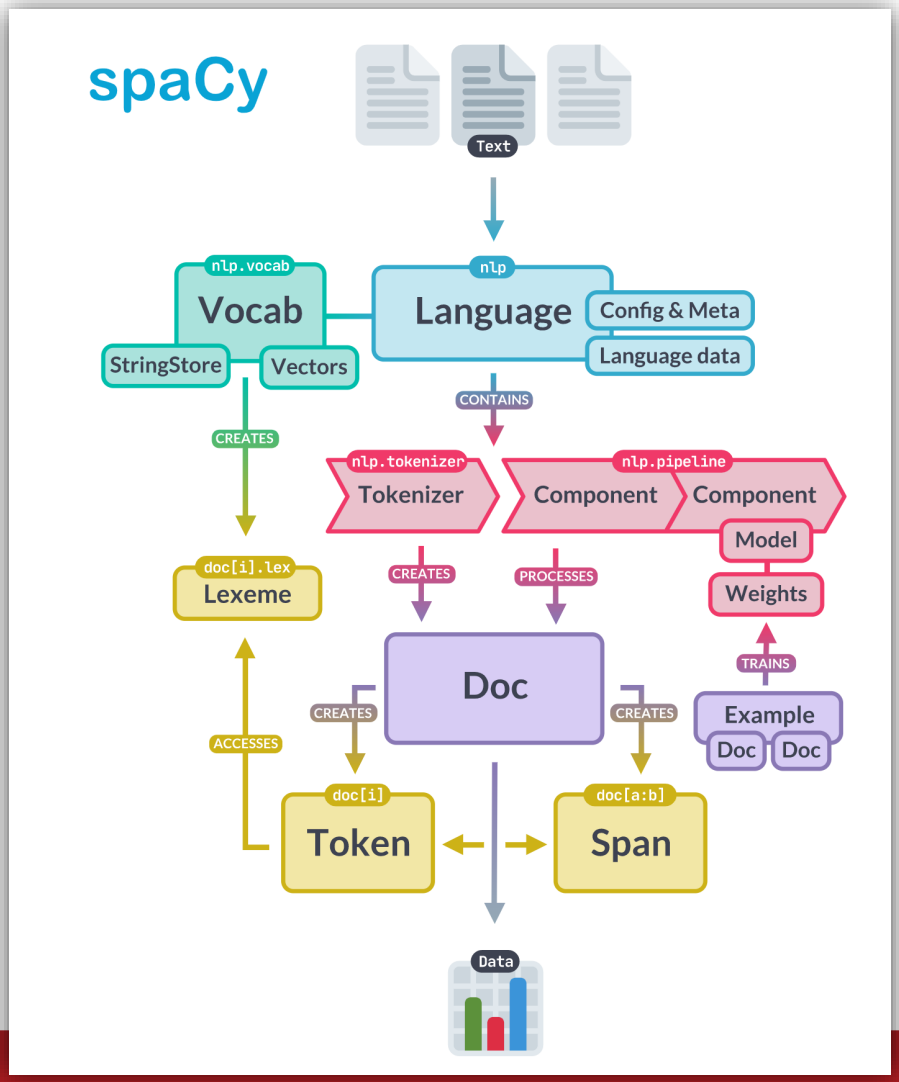
doc_ent = nlp("When Sebastian Thrun started working on self-driving cars at Google "
              "in 2007, few people outside of the company took him seriously.")
displacy.serve(doc_ent, style="ent")
```



https://spacy.io/usage/spacy-101

Prof. Dr. Hamed Shariat Yazdi
Department of Computer Science
Darmstadt Univeristy of Applied Sciences

spaCy architecture overview



Agenda

- Overview
- spaCy
- Simple question answering
- Quick check

Task: answer questions about the movie „A Beautiful Mind“

A Beautiful Mind is a 2001 American biographical drama film based on the life of the American mathematician John Nash, a Nobel Laureate in Economics and Abel Prize winner. The film was directed by Ron Howard, from a screenplay written by Akiva Goldsman. It was inspired by the bestselling, Pulitzer Prize-nominated 1998 book of the same name by Sylvia Nasar. The film stars Russell Crowe, along with Ed Harris, Jennifer Connelly, Paul Bettany, Adam Goldberg, Judd Hirsch, Josh Lucas, Anthony Rapp, and Christopher Plummer in supporting roles. The story begins in Nash's days as a graduate student at Princeton University. Early in the film, Nash begins to develop paranoid schizophrenia and endures delusional episodes while watching the burden his condition brings on his wife Alicia and friends.

Example

Example question



Text basis for answer

[…] based on the life of the American mathematician John Nash, a Nobel Laureate in Economics and Abel Prize winner.

Prof. Dr. Hamed Shariat Yazdi
Department of Computer Science
Darmstadt Univeristy of Applied Sciences

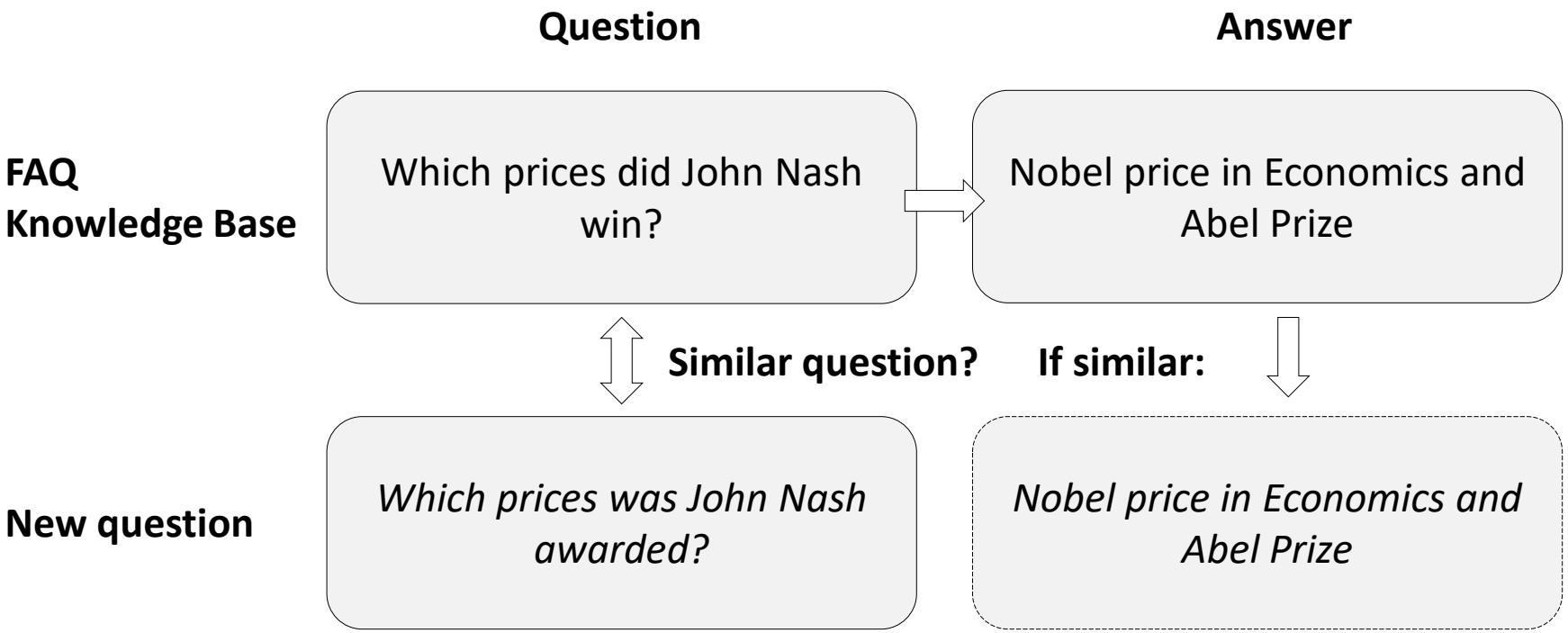
https://upload.wikimedia.org/wikipedia/commons/5/5c/Amazon_Echo.jpg

Simple QA:
FAQ as a knowledge base (training data)

	A	B
1	**Question**	**Answer**
2	From which country is the film A Beautiful Mind?	America
3	When was the film A beautiful Mind released?	2001
4	Of which genre is the film A Beautiful Mind?	biographical drama
5	What is John Nash's profession?	mathematician
6	Which prices did John Nash win?	Nobel price in Economics and Abel Prize
7	By which book was the film inspired?	the bestselling, Pulitzer Prize-nominated 1998 book of the same name by Sylvia Nasar
8	Who is acting in the film?	Russell Crowe, along with Ed Harris, Jennifer Connelly, Paul Bettany, Adam Goldberg, Judd Hirsch, Josh Lucas, Anthony Rapp, and Christopher Plummer in supporting roles
9	At which university was Nash a graduate student?	Princeton University
10	What disease did Nash develop?	paranoid schizophrenia
11	When did the film open in the United states?	December 21, 2001
12	How much did the film gross?	over $313 million worldwide
13	How many Academy Awards did the film win?	four
14	Did the film win an Academy Award?	yes

Prof. Dr. Hamed Shariat Yazdi
Department of Computer Science
Darmstadt Univeristy of Applied Sciences

Task in simple QA: Find answers to similar questions



Test set:
New questions to be answered

	Question	Answer
2	From which country is the film A Beautiful Mind?	America
3	Where was the film A Beautiful Mind made?	America
4	In which country was the film A Beautiful Mind made?	America
5	Where was the movie A Beautiful Mind made?	America
6	From which country is the film?	America
7	From which country is this film?	America
8	When was the film A beautiful Mind released?	2001
9	Of which genre is the film A Beautiful Mind?	biographical drama
10	Which kind of film is A Beautiful Mind?	biographical drama
11	Which type of movie is A Beautiful Mind?	biographical drama
12	What is John Nash's profession?	mathematician
13	What is John Nash's job?	mathematician
14	Which prices did John Nash win?	Nobel price in Economics and Abel Prize
15	Which awards did John Nash win?	Nobel price in Economics and Abel Prize
16	Which prices was John Nash awarded?	Nobel price in Economics and Abel Prize
17	By which book was the film inspired?	the bestselling Pulitzer Prize-nominated 1998 book of the same name by Sylvia Nasar
18	Which book is the basis of the film?	the bestselling Pulitzer Prize-nominated 1998 book of the same name by Sylvia Nasar
19	Who is acting in the film?	Russell Crowe, along with Ed Harris, Jennifer Connelly, Paul Bettany, Adam Goldberg, Judd Hirsch, Josh Lucas, Anthony Rapp, and Christopher Plummer in supporting roles
20	Who are the actors in the film?	Russell Crowe, along with Ed Harris, Jennifer Connelly, Paul Bettany, Adam Goldberg, Judd Hirsch, Josh Lucas, Anthony Rapp, and Christopher Plummer in supporting roles
21	Who is starring in this film?	Russell Crowe, along with Ed Harris, Jennifer Connelly, Paul Bettany, Adam Goldberg, Judd Hirsch, Josh Lucas, Anthony Rapp, and Christopher Plummer in supporting roles
22	Who is acting in the movie?	Russell Crowe, along with Ed Harris, Jennifer Connelly, Paul Bettany, Adam Goldberg, Judd Hirsch, Josh Lucas, Anthony Rapp, and Christopher Plummer in supporting roles
23	At which university was Nash a graduate student?	Princeton University
24	Where did John Nash graduate?	Princeton University
25	Where did John Nash get his university degree?	Princeton University
26	What disease did Nash develop?	paranoid schizophrenia
27	What illness did John Nash have?	paranoid schizophrenia
28	Which disorder did Nash suffer from?	paranoid schizophrenia

Loading data

```
import pandas as pd
```

```
train_data = pd.read_csv('data/QA_train_data.csv')
test_data = pd.read_csv('data/QA_test_data.csv')
```

Prediction logic

```
def predict_answer(question: str)-> str:
    """
    Predicts an answer to a given question
    :param question: question string
    :return: answer string or None if insufficient confidence
    """
    # Your code…
```

Predicting

```
# Predict
predicted = test_data['Question'].apply(lambda q: predict_answer(q))
actual = test_data['Answer']
```

Evaluating results

```
# Evaluate
score = evaluate_results(predicted, actual)
```

```
def evaluate_result(predicted: str, actual: str) -> int:
    """
    Evaluates individual string answer string pair
    :param predicted: answer predicted by predictor
    :param actual: correct answer according to data set
    :return: +1 if predicted answer is correct,
    0 if no answer is predicted (None),
    -1 if answer is wrong
    """
    if predicted is None:
        return 0
    elif predicted == actual:
        return 1
    else:
        return -1
```

```
def evaluate_results(predicted: pd.Series, actual: pd.Series) -> float:
    """
    Evaluates a series of answer pairs
    :param predicted: Series of predicted answers
    :param actual: Series of correct answers
    :return: Float value between -1 (worst) and +1 (best)
    """
    sum = 0
    for index, value in predicted.items():
        eval = evaluate_result(value, actual[index])
        sum += eval
    result = sum / predicted.size
    return result
```

Ideas for implementing QA?



Agenda

- Overview
- spaCy
- Simple question answering
- Quick check

Quick check
NLP – question answering

1. What is the task of question answering?
2. Explain functionality of spaCy